

synth-attest: An Open, Vendor-Neutral Verifiable-Identity and Tamper-Evident Audit Layer for DNA-Synthesis Order Compliance

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Abstract—DNA-synthesis screening answers whether a sequence is dangerous, but not who ordered it, whether they are a verified researcher, or whether a trustworthy record of the decision exists. As regulation (the proposed US Biosecurity Modernization and Innovation Act, the US OSTP framework, UK DSIT guidance, and the proposed EU Biotech Act) converges on mandating customer identity verification and recordkeeping, no shared, vendor-neutral tooling exists; commercial entrants use centralized, per-provider Know-Your-Customer (KYC) workflows. We present synth-attest, an open-source layer that sits on top of (and never replaces) sequence screening, adding three capabilities built on W3C standards: a portable researcher-identity credential, a provider-issued revocable exemption credential, and a privacy-preserving, tamper-evident order-audit record. The design is vendor-neutral by construction: the core has zero proprietary dependencies and runs on a dependency-free reference engine, with any W3C Verifiable Credential backend pluggable behind a stable interface. We report a working prototype (592 lines of core Python, 42 tests) validated in continuous integration on Python 3.11 and 3.12 in two configurations: a no-proprietary-dependency core (32 passed, 10 backend tests skipped) and an optional backend validated source-blind against its published package (42 passed). We state the threat model honestly, scoping the contribution to fraud, attribution, and deterrence at the acquisition node.

Index Terms—biosecurity, DNA synthesis, verifiable credentials, decentralized identifiers, audit trails, know-your-customer, compliance

I. INTRODUCTION

Advances in machine learning lower the barrier to designing biological sequences of concern. A 2025 study found that AI-designed variants of toxic proteins evaded the sequence-screening tools used by synthesis providers, with one tool missing more than 75% of roughly 76,000 variants before patching [1]. This makes *who is ordering* as important as *what is ordered*, and the synthesis-order step one of the few remaining physical chokepoints for governance. Sequence-screening systems such as SecureDNA [3] and the IBBIS Common Mechanism [4] assess whether an ordered sequence matches a hazard, but they do not establish whether the customer is a verified and legitimate researcher, or whether the order decision leaves a tamper-evident, independently verifiable record. Recent and proposed regulation, including the

US Biosecurity Modernization and Innovation Act of 2026 [5] (a bill), the US OSTP nucleic-acid-synthesis framework, UK DSIT screening guidance [6] (in force), and the proposed EU Biotech Act, pushes providers toward customer identity verification and recordkeeping. The 2026 policy literature converges on identity as the durable control [2], but the available implementations are closed and centralized: commercial entrants such as Aclid offer per-provider KYC software, and no open-source, decentralized implementation with a tamper-evident cross-provider order audit exists.

We present **synth-attest**, an open, vendor-neutral layer that adds the missing identity and audit capabilities on top of existing screening. We do not claim to have discovered that identity matters; we provide the open, decentralized, audited implementation the 2026 literature calls for but has not built. It is deliberately scoped: it does not screen sequences and never handles sequence data.

II. THREAT MODEL AND HONEST SCOPE

We map the engineered-pandemic kill chain as: capability uplift, acquire materials, synthesize, weaponize, release. synth-attest intervenes at a single node, *acquire materials*, and only for actors routing through compliant providers. It addresses three failure modes there: identity fraud (spoofed or unverified customers), weak attribution (no portable record of who ordered what), and low deterrence. It explicitly does *not* address capability uplift, benchtop or non-compliant synthesis, weaponization, or a genuinely credentialed malicious insider; identity proves identity, not intent. Any claim that the layer “prevents engineered pandemics” is unsupported and avoided.

III. DESIGN

synth-attest provides three artifacts and one privacy property (Fig. 1).

A. Portable researcher-identity credential

A W3C Verifiable Credential [7] bound to a Decentralized Identifier [8] asserts a verified researcher and institutional affiliation only. It is explicitly not a clearance, and carries

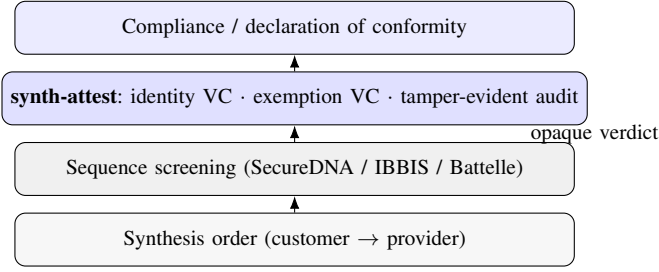


Fig. 1. Layered architecture. synth-attest adds an identity and audit layer above sequence screening, which it never replaces. Only an opaque verdict (cleared / flagged / tier-review) crosses the boundary; no sequence data enters the layer.

no risk-tiering field. Portability lets a researcher avoid re-onboarding at every provider, addressing the documented gap that customer screening is currently re-performed per provider and rarely shared.

B. Provider-issued exemption credential

Following existing exemption-certification practice, an exemption credential is issued by the provider (which retains regulatory liability) to authorize a sequence *class*, never a sequence. It is time-bound and revocable. This separates the portable identity (about the person) from the order-specific authorization (the provider’s local decision), correcting a common conflation between credentialing a person and clearing an order.

C. Privacy-preserving tamper-evident audit

Each order decision produces an append-only, signed record held by the provider. Only a salted SHA-256 commitment over (salt, order id, verdict, customer DID) is anchored, and commitments are anchored in fixed-size batches of $B = 16$, padded with indistinguishable random fillers and sorted, so that order volume, timing, and the screening verdict are not recoverable from what is anchored. A customer-facing receipt never contains the verdict, preventing the audit channel from becoming a screening-evasion oracle.

D. Infohazard guard

No sequence-like data may enter the layer. A guard rejects any payload containing a nucleotide-alphabet ($[ACGTU]$) run of length ≥ 20 , erring toward false positives, the safe direction for an infohazard boundary.

IV. VENDOR NEUTRALITY

The core has zero proprietary dependencies. A dependency-free reference engine (`StubEngine`) implements the full contract (issue, verify, detect tampering, revoke, selectively disclose), so the entire flow runs with only the Python standard library. A swappable `CredentialEngine` interface allows any conformant W3C VC/DID backend; an optional adapter (`AttestixEngine`) is provided but not required, and auto-detects the backend’s service or functional surface. Continuous integration enforces independence: one job installs the core,

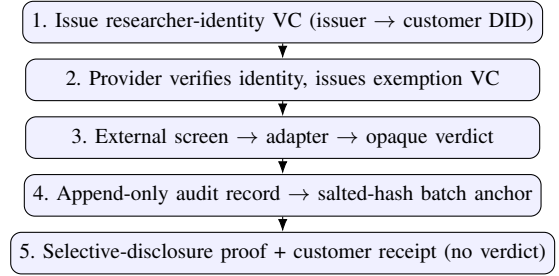


Fig. 2. End-to-end order flow, implemented as the project demo and runnable on both engines.

TABLE I
ARTIFACTS AND THEIR KEY PROPERTIES

Artifact	Issuer	Key property
Researcher identity VC	registry	portable; identity only, no tier
Exemption VC	provider	sequence- <i>class</i> , time-bound, revocable
Order audit record	provider	salted SHA-256, batch $B=16$ anchor
Disclosure proof	holder	selective; reveals chosen fields only

asserts the optional backend is absent, and runs the suite; a second installs the optional backend and validates it source-blind against the published package.

V. IMPLEMENTATION AND VALIDATION

The prototype is 592 lines of core Python across five modules (engine, schemas, audit, adapters, custody), plus reference adapters that map SecureDNA-shaped and IBBIS-Common-Mechanism-shaped screening results to the three opaque verdicts, and a custodial registry so researchers need not operate a key wallet (a provider or neutral registry custodies the credential, keyed by a salted hash of an identifier). Every credential and audit field is traced to a named regulatory obligation, each marked verified or unverified, and a test fails if any schema field lacks a mapping row. Table I summarizes the artifacts; Table II reports validation. The suite is 42 tests; in the no-proprietary-dependency configuration 32 pass and the 10 backend-specific tests skip, and with the optional backend active (validated source-blind against the published package) all 42 pass. Both configurations are green in CI on Python 3.11 and 3.12. The suite includes hardening tests for the infohazard guard against obfuscated sequences, commitment domain separation, and custody pseudonymization.

Three properties are demonstrated by dedicated tests rather than asserted: (i) tampering with any credential field causes verification to fail; (ii) a revoked credential fails verification and cannot be presented; and (iii) the anchored batch root has fixed width and does not reveal the number of real orders in the batch, so order volume is not recoverable from what is anchored.

VI. RELATED WORK AND POSITIONING

Sequence-screening systems [3], [4] are complementary, not competing; synth-attest consumes their verdicts as opaque

TABLE II
VALIDATION RESULTS (CONTINUOUS INTEGRATION)

Configuration	Passed	Skipped
Core only (no proprietary dependency)	32	10
Optional backend (source-blind, published)	42	0
Python 3.11 and 3.12; 42 tests collected; 592 LOC core.		

tokens. Commercial KYC entrants such as Aclid use centralized, per-provider identity verification with no portable customer-held credential or cross-provider audit. SecureDNA’s Exemption Certification System pre-vets customers and orders but is not a portable identity. IBBIS has published customer-screening forms and plans a 2026 identity landscape analysis but has not built credentials. The “Know Your Scientist” proposal [2] governs access to biological AI design tools (a different chokepoint) via institution-vouching and shared exclusion lists, and explicitly does not use verifiable credentials, decentralized identifiers, or a cryptographic audit. Against this landscape, the distinguishing and best-supported property of synth-attest is being the only *open-source, decentralized W3C-credential implementation with a tamper-evident order audit*; portability is a feature, not the core novelty.

VII. LIMITATIONS AND FUTURE WORK

This is a prototype, not a production system, and the reported results are functional and integration tests, not an empirical study of deployed effectiveness. Four limitations are explicit. (1) *Privacy of the audit*: fixed-size batches hide the order count within a batch, but the number of batches and their timing still leak coarse volume; closing this needs fixed-cadence anchoring. (2) *Durability*: the reference implementation holds records and anchors in memory, so tamper-evidence does not survive a restart; persistence is the first production step. (3) *Data protection*: storing researcher identity attributes creates a tension between an immutable audit and the right to erasure under GDPR/DPDP-style regimes; the intended mitigation is to keep personal data off the anchored ledger and anchor only hashes. (4) *Dependency maturity*: the optional backend is a release candidate with limited throughput, unsuitable for high order volumes as-is. A credential-tiering field is deliberately omitted pending confirmation that any real framework tiers researcher clearance rather than data sensitivity. An issuer trust-root design (accrediting identity issuers without recreating a centralized authority) is the principal open architectural question. A provider using synth-attest only partially meets its obligations, since the layer adds identity and audit but not screening; we therefore describe it as helping to satisfy, not satisfying, these requirements.

VIII. CONCLUSION

synth-attest supplies the identity and audit layer that sequence screening leaves open, as an honest, narrowly scoped, and vendor-neutral open-source contribution to nucleic-acid synthesis governance.

APPENDIX A REPRODUCIBILITY OF REPORTED FIGURES

All numeric claims are mechanically checkable from the public repository at the cited commit: 592 LOC (`wc -l src/synth_attest/*.py`); 42 tests collected (`pytest --co`); 32 passed / 10 skipped core and 42 passed with the optional backend (`pytest`); batch size $B=16$ and the SHA-256 commitment (`src/synth_attest/audit.py`); the length-20 [ACGTU] guard (`src/synth_attest/schemas.py`); optional backend version pinned in `pyproject.toml`. The two CI configurations are defined in `.github/workflows/ci.yml`.

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